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SemanticDistance: Compute Semantic Relatedness between Constituents of Sequential Continuous Text

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Summary

`SemanticDistance` computes the pairwise distance between units of language (e.g., word-to-word, ngram-to-word, turn-to-turn) in both structured language samples and unstructured word lists. `SemanticDistance` has cleaning and formatting options including stopword removal and lemmatization. The package computes two complementary cosine distance indices for each pairwise contrast of interest. `SemanticDistance` can also be used to visualize relationships between words. Visualization options include hierarchical cluster dendrograms and igraph network plots.

Statement of need

There are many compelling theoretical and clinical applications for measuring conceptual similarity between words and larger chunks of language (e.g., detection of subtle semantic impairments based on naturalistic language sampling, designing controlled word pair stimuli for language experiments). Although word embeddings such as `Word2Vec` (Mikolov et al. 2013), `LSA` (Landauer et al. 1998), and `GloVE` (Pennington et al. 2014) are widely available, `SemanticDistance` is unique in its capacity to compute rolling measures of semantic distance (e.g., one word to the previous word in continuous text) using qualitatively different sources (e.g., experiential norms vs. embeddings). `SemanticDistance` also offers in text cleaning and data visualization options such as cluster dendrograms and network plots.

State of the field

Many of the text cleaning, formatting, and pairwise distance operations within `SemanticDistance` are redundant with other R packages such as `embedplyr` (Teitelbaum 2025), `quanteda` (Benoit et al. 2018), `PsychWordVec` (Bao 2026), and `text` (Kjell et al. 2023). Principal advantages of `SemanticDistance` over existing packages include:

1) An integrated text cleaning, distance computation, annd network workflow that is exceuted within the same package. 2) A coding argument structure that is simple/basic enough so that the package will be accessible to non-experts in computational linguistics and/or natural language processing. 3) Computation of rolling metrics of semantic distance, allowing for analyses of word-to-word shifts in meaning in native prose.

Software design

`SemanticDistance` is an open-source R package designed to accommodate a wide range of user expertise (e.g., neuroscience, linguistics, computer science). The software does not leverage artificial intelligence or use any proprietary dependencies. It instead derives mathematical indices of semantic distance between constituents (e.g., words, ngrams) by extracting their respective vectors from two static lookup datasets. The first dataset contains GloVE-derived word embeddings for many English words. The second dataset contains experiential semantic norms for sensory, motor, and emotion salience (e.g., color, sound, motor).

Research impact statement

`SemanticDistance` has supported one peer-reviewed publication to date in the journal`Cortex` (Mechtenberg et al. 2026) along with one refereed conference presentation at the annual conference of the Society for the Neurobiology of Language. One novel advance of `SemanticDistance` is its capacity to model rolling semantic distance in continuous text by transforming each constituent's distance to the previous word or ngram to a continuous time series.

Description

Semantic distance is an empirical measure of conceptual relatedness between two or more words situated within an n-dimensional space (Reilly et al. 2025). Semantic spaces differ in their dimensionality and biological plausibility. As an R package, `SemanticDistance` bundles text cleaning and formatting options (e.g., stopword removal, lemmatization) with two complementary semantic distance metrics that characterize/quantify the similarity between pairs of language constituents (e.g., words, ngrams, turns). `CosDist_Glo` reflects cosine distance between vectors derived from a GloVe word embedding model (Pennington et al. 2014). `CosDist_SD15` refects cosine distances derived from a 15 dimension space comprised of meaningful psychological constructs spanning motor, emotion, and action variables (Reilly et al. 2023). Although there are many mathematical approaches to aggregating vectors across words/documents (Teitelbaum and Simchon 2025), `SemanticDistance` computes arithmetic means of multi-word vectors prior to deriving cosine distances between constituents.

`SemanticDistance` computes indices of relatedness between lexical constituents within the following text formats:
1: **monologues**: stories, narratives, and structured text
2: **dialogues**: turn-to-turn within two-person conversation transcripts.

3: **word pairs in columns**: word pairs
4: **unordered lists**: bags-of-words

One of the most innovative functions of `SemanticDistance` is its capacity to generate rolling measures of semantic distance within stories or narratives with chunk sizes (e.g., word-to-word, turn-to-turn). Chunk size is an optional argument for many of the software's functions. For example, if a user specified a chunk or n-gram size of 5 words, the software would automatically compute a rolling measure of semantic distance between each new word in a language sample relative to the five words that preceded it. One consideration of applying `SemanticDistance` to naturalistic language samples relates to coverage of contemporary English. `SemanticDistance` relies on an internal lookup dataset populated with semantic vectors for >60k frequent and familiar English words.

Users should exercise caution regarding potential bias from missing observations (NAs) (e.g., proper nouns, low frequency or obscure words), especially in small corpora.

We envision a number of potential use cases for the `SemanticDistance` package as applied to naturalistic discourse. For example, a researcher interested in the effects of Alzheimer's Disease on semantic cohesion between words could model running measures of distance across entire language transcripts, allowing a glimpse into how cohesion changes over time. Other applications involve dissociations in semantic distance type (i.e., experiential vs. embedding) and ngram kernel size (e.g., rolling distance from larger chunks of words to each new word). Other potential uses include matching stimuli for relatedness in language experiments.

We invite the user community to contribute suggestions for functionality along with code using the package's bug-reporting notification feature (see ReadMe) or by contacting the package maintainer (Jamie Reilly) directly.

AI usage disclosure

`SemanticDistance` uses an internal database composed of many words with a fixed set of embeddings. The software does not use AI or rely on proprietary package dependencies. We did use GPT 4.0 (OpenAI) to troubleshoot R coding errors and assist with generating complex regular expressions (regex) used for cleaning text data.

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